

Evolution and Generations of Computers

Learning Outcomes

- Name all 5 generations of computers in correct chronological order with their year ranges.
- Identify and explain the key electronic technology used in each generation.
- Compare all 5 generations on the basis of size, speed, heat, cost, and example machines.
- Explain, with reasoning, why computers became smaller, faster, and cheaper over time.
- Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full explanations and diagrams.

1. What is a “Generation” of Computers?

In the context of computer history, a 'generation' refers to a distinct period of technological development, defined primarily by the main electronic component or technology used to build the central processing circuitry of computers during that time. Historians and computer scientists commonly divide the history of electronic computers into five such generations.

A very important pattern to notice, and to mention in your exam answers, is that with each successive generation, computers became simultaneously smaller in physical size, faster in processing speed, lower in manufacturing cost, more reliable, and more energy-efficient (producing less heat and consuming less power) than the generation before it. This consistent trend of improvement is one of the most remarkable stories in the history of technology.

2. First Generation (1940–1956): Vacuum Tubes

The very first electronic computers, built between approximately 1940 and 1956, relied on vacuum tubes as their basic electronic switching component. A vacuum tube is a fragile, glass-enclosed device, somewhat similar in physical appearance to an old-style incandescent light bulb, which controls the flow of electric current and can act as a switch — the basic building block needed to represent and process binary information (0s and 1s).

Computers of this generation shared several characteristic drawbacks. They were physically enormous, often occupying an entire large room; they consumed massive amounts of electrical power; they generated tremendous amounts of heat, requiring dedicated cooling arrangements; and vacuum tubes themselves were notoriously unreliable, burning out frequently (much like a light bulb does) and requiring constant replacement and maintenance by dedicated teams of engineers. Programming these machines was also extremely difficult, often requiring the physical rewiring of circuits for each new task.

Example Machine: ENIAC (Electronic Numerical Integrator and Computer), completed in 1945, is one of the most famous examples of a first-generation computer. It occupied an entire room, weighed approximately 30 tons, contained roughly 18,000 vacuum tubes, and required a dedicated team of engineers just to keep it operational.

- **First Generation (1940 to 1956): Using Vacuum Tubes**

- **Hardware Technology:**

- » The first generation of **computers used vacuum tubes** for circuitry and magnetic drums for memory. The **input** to the computer was through **punched cards and paper tapes**. The output was displayed as printouts.

First Generation Computers



Vacuum Tubes

- **First Generation (1940 to 1956): Using Vacuum Tubes**

- **Software Technology :**
 - » The instructions were written in **machine language**. Machine language uses 0s and 1s for coding of the instructions. The first generation computers could solve one problem at a time.
 - **Computing Characteristics:**
 - » The computation time was **in milliseconds**.
 - **Physical Appearance:**
 - » These computers were **enormous in size** and required a large room for installation.

- **First Generation (1940 to 1956): Using Vacuum Tubes**

- **Application:**
 - » They were used for **scientific applications** as they were the fastest computing device of their time.
 - **Examples:**
 - » UNIVersal Automatic Computer (**UNIVAC**),
 - » Electronic Numerical Integrator And Calculator (**ENIAC**),
 - » Electronic Discrete Variable Automatic Computer (**EDVAC**)

- **Disadvantages of 1st Generation:**

- The first generation computers used a large number of vacuum tubes and thus **generated a lot of heat**.
 - They consumed a great deal of electricity and were **expensive to operate**.
 - The machines were prone to frequent malfunctioning and **required constant maintenance**.
 - Since first generation computers used machine language, they were **difficult to program**.

3. Second Generation (1956–1963): Transistors

The second generation of computers, spanning roughly 1956 to 1963, was defined by the replacement of the bulky, fragile, heat-generating vacuum tube with a newly invented device called the transistor. A transistor performs essentially the same electronic switching function as a vacuum tube, but is dramatically smaller (often no larger than a small pea or button), far more reliable, consumes significantly less electrical power, and generates far less heat.

This single technological shift allowed second-generation computers to become considerably smaller (though still often cupboard- or room-sized by today's standards), faster, more energy-efficient, and considerably more reliable than their first-generation predecessors. It also made computers more commercially practical, as they became somewhat more affordable and manageable for large organisations to purchase and maintain.

Analogy: This transition is very similar to the historical shift from bulky, valve-based (vacuum-tube) radio sets to small, battery-powered transistor radios — the same fundamental idea of replacing a large, fragile, power-hungry component with a small, robust, efficient one, applied here to computers instead of radios.

Example Machine: The IBM 1401 is a well-known example of a commercially successful second-generation, transistor-based computer, widely used by businesses for data processing tasks.

• Second Generation (1956 to 1963): Using Transistors

• Hardware Technology :

Second Generation of Computers



- » Transistors replaced the vacuum tubes of the first generation of computers.
- » Transistors allowed computers to become smaller, faster, cheaper, energy efficient and reliable.
- » The second generation computers used magnetic core technology for primary memory. They used magnetic tapes and magnetic disks for secondary storage.
- » The input was still through punched cards and the output using printouts.
- » They used the concept of a stored program, where instructions were stored in the memory of computer.

• Second Generation (1956 to 1963): Using Transistors

• Software Technology:

- » The instructions were written using the assembly language.
- » Assembly language uses mnemonics like ADD for addition and SUB for subtraction for coding of the instructions.
- » High-level programming languages, such as early versions of COBOL and FORTRAN were also developed during this period

• Second Generation (1956 to 1963): Using Transistors

• Computing Characteristics :

- » The computation time was in microseconds.

• Physical Appearance :

- » Transistors are smaller in size compared to vacuum tubes, thus, the size of the computer was also reduced.

• Examples: PDP-8, IBM 1401 and CDC 1604

- The cost of commercial production of these computers was very high, though less than the first generation computers. The transistors had to be assembled manually in second generation computers

4. Third Generation (1964–1971): Integrated Circuits (IC)

The third generation, spanning roughly 1964 to 1971, was made possible by a further technological leap: the invention of the Integrated Circuit (IC). An integrated circuit is a single, small piece of semiconductor material (typically silicon) onto which many individual transistors, along with other necessary electronic components, are manufactured together and interconnected as a single compact unit.

This innovation allowed engineers to pack a very large number of transistors, which had previously needed to be wired together individually and by hand, onto one single small chip. The result was a further dramatic reduction in the physical size of computers, alongside significant improvements in processing speed, reliability, and manufacturing cost. For the first time, computers became genuinely practical purchases for ordinary businesses, banks, and universities, rather than being restricted mainly to large government research laboratories and major corporations.

Example Machine: The IBM System/360 is a well-known and highly influential example of a third-generation computer, widely adopted across many industries.

• Third Generation (1964 to 1971): Using Integrated Circuits:

• Hardware Technology:

- » The third generation computers used the Integrated Circuit (IC) chips.
- » In an IC chip, multiple transistors are placed on a silicon chip.
- » The use of IC chip increased the speed and the efficiency of computer, manifold.
- » The keyboard and monitor were used to interact with the third generation computer, instead of the punched card and printouts.

• Third Generation (1964 to 1971): Using Integrated Circuits:

• Physical Appearance :

- » The size of these computers was quite small compared to the second generation computers.

• Examples: IBM 370, PDP 11.

- The third generation computers used less power and generated less heat than the second generation computers. The cost of the computer reduced significantly, as individual components of the computer were not required to be assembled manually. The maintenance cost of the computers was also less compared to their predecessors.

5. Fourth Generation (1971–Present): Microprocessors

The fourth generation, beginning around 1971 and continuing to the present day, was defined by an even more remarkable advance: the invention of the microprocessor. A microprocessor is essentially an entire Central Processing Unit (CPU) — the complete 'brain' of a computer — manufactured onto a single, tiny integrated circuit chip.

This development is arguably the single most important turning point in the entire history of computing, because it made possible the creation of small, genuinely affordable Personal Computers (PCs) that ordinary individuals, small businesses, and schools could purchase and use, rather than computing remaining the exclusive domain of large organisations. Over the following decades, this same microprocessor technology continued to shrink in size and grow in power, eventually enabling laptops, and later, smartphones — which are themselves extremely powerful microprocessor-based computers small enough to fit in a pocket.

Example: Almost every modern computer in everyday use today, including the desktop computers in your college laboratory and the smartphone in your own pocket, belongs to this fourth generation, being built around microprocessor technology.

- **Fourth Generation (1971 to present): Using Microprocessors**

- Hardware Technology:
 - » They use the **Large Scale Integration (LSI) and the Very Large Scale Integration (VLSI) technology**. Thousands of transistors are integrated on a small silicon chip using LSI technology. VLSI allows hundreds of thousands of components to be integrated in a small chip.
 - » **Microprocessor is a chip containing millions of transistors** and components, and, designed using LSI and VLSI technology.
 - » **Semiconductor memory** replaced the earlier magnetic core memory, resulting in fast random access to memory.
 - » Secondary storage device like magnetic disks became smaller in physical size and larger in capacity.

- **Fourth Generation (1971 to present): Using Microprocessors**

- » The linking of computers is another key development of this era. The computers were linked to form networks that **led to the emergence of the Internet**. This generation also saw the development of pointing devices like mouse, and handheld devices.
- Software Technology:
 - » Several **new operating systems like the MS-DOS and MS-Windows developed during this time**. This generation of computers supported Graphical User Interface (GUI).
 - » GUI is a user-friendly interface that allows user to interact with the computer via menus and icons.
 - » High-level programming languages are used for the writing of programs.

- **Fourth Generation (1971 to present): Using Microprocessors**

- Computing Characteristics:
 - » The computation time is in **picoseconds**.
- Physical Appearance :
 - » They are smaller than the computers of the previous generation.
- **Ted Hoff** produced the first microprocessor in 1971 for **Intel**. It was known as Intel 4004.

6. Fifth Generation (Present–Future): Artificial Intelligence

The fifth generation, encompassing the present day and extending into the anticipated future, is somewhat different from the previous four generations, because it is defined less by a single new physical electronic component, and more by a new capability: Artificial Intelligence (AI). Fifth-generation computing systems are designed with the goal of recognising speech and natural human language, interpreting images, identifying complex patterns in large amounts of data, and — crucially — 'learning' from data and experience in order to improve their own performance over time, rather than simply executing a completely fixed, unchanging set of instructions.

Examples of fifth-generation ideas in everyday use include voice assistants on smartphones (which can understand and respond to spoken natural language), face-unlock systems (which use image recognition to identify a specific person), and AI-based chatbots (which can hold natural, flowing conversations with users).

• **Fifth Generation (Present and Next): Using Artificial Intelligence:**

- The goal of fifth generation computing is to develop computers that **are capable of learning and self-organization**.
- The fifth generation computers use **Super Large Scale Integrated (SLSI)** chips that are able to store millions of components on a single chip.
- These computers have large memory requirements.
- This generation of computers **uses parallel processing** that allows several instructions to be executed in parallel, instead of serial execution. Parallel processing results in faster processing speed.

• **Fifth Generation (Present and Next): Using Artificial Intelligence:**

- **Quantum computation and molecular and nanotechnology** will radically change the face of computers in years to come.
- The goal of fifth-generation computing is to develop devices that respond to **natural language input** and are capable of learning and self-organization.

Important Clarification for Exams: A computer is not automatically classified as “fifth generation” simply because it was recently manufactured or purchased. It qualifies as fifth generation specifically when it is built around genuine Artificial Intelligence capability. In fact, most everyday laptops and desktop computers sold today are still fundamentally fourth-generation machines, built around conventional microprocessor technology, even though they may be brand new.

5 Generations — Computers Got Smaller & Faster

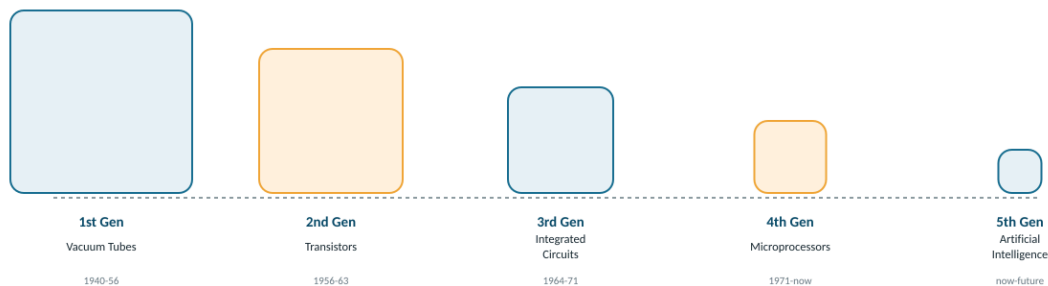


Fig 3.1 — The 5 Generations: Computers Got Smaller and Faster Over Time

7. Comparison of All 5 Generations

Generation	Technology	Size	Speed	Example
1st (1940-56)	Vacuum Tubes	Room-sized	Slow	ENIAC
2nd (1956-63)	Transistors	Cupboard-sized	Faster	IBM 1401
3rd (1964-71)	Integrated Circuits	Desk-sized	Much faster	IBM 360
4th (1971-present)	Microprocessors	Laptop/palm-sized	Very fast	Personal Computers
5th (present-future)	Artificial Intelligence	Palm-sized & smaller	Extremely fast	AI-enabled smartphones

8. Fill in the Blanks

1. The 1st generation of computers (1940-1956) used _____ as their main electronic component.
2. The 2nd generation of computers used _____, which were smaller and more reliable than vacuum tubes.

- 3. An _____ is a single chip containing many transistors combined together, used in 3rd generation computers.
- 4. A _____ is an entire CPU built onto a single chip, defining the 4th generation.
- 5. The 5th generation of computers focuses mainly on _____.
- 6. ENIAC is a famous example of a _____ generation computer.
- 7. The IBM System/360 is an example of a _____ generation computer.
- 8. With each new generation, computers became smaller, faster, and _____ in cost.
- 9. Voice assistants and face-unlock are everyday examples of _____ generation ideas.
- 10. Most laptops sold today are still primarily _____ generation machines, based on microprocessors.

Answers: 1. Vacuum Tubes | 2. Transistors | 3. Integrated Circuit (IC) | 4. Microprocessor | 5. Artificial Intelligence | 6. 1st | 7. 3rd | 8. cheaper / lower | 9. 5th | 10. 4th

9. True or False

- 1. First generation computers were small enough to fit on a desk.
- 2. Transistors are smaller and more reliable than vacuum tubes.
- 3. An Integrated Circuit (IC) contains many transistors combined onto a single chip.
- 4. The microprocessor made personal computers (PCs) possible.
- 5. Any brand-new laptop automatically counts as a 5th generation computer.
- 6. 5th generation computing is mainly about being able to recognise speech, images, and patterns, and learning from data.
- 7. Computers generally became larger and more expensive with each new generation.

Answers with Explanation

- 1. False.** First generation computers, such as ENIAC, were room-sized and extremely bulky.
- 2. True.** This is exactly why 2nd generation computers, built with transistors, were smaller and more dependable than 1st generation vacuum-tube machines.
- 3. True.** This is the defining innovation of the 3rd generation of computers.
- 4. True.** By placing an entire CPU on a single chip, the microprocessor made small, affordable personal computers achievable for the 4th generation.
- 5. False.** A device is 5th generation only if it is built around genuine Artificial Intelligence capability; most current laptops are still 4th-generation, microprocessor-based machines.
- 6. True.** This is the defining characteristic of 5th generation, Artificial Intelligence-based computing.
- 7. False.** The opposite is true — computers became smaller, faster, and cheaper with each successive generation.

10. Short Answer Questions (2–3 Marks)

- Which technology was used in 1st generation computers? (2 marks)
- Name an example machine of the 3rd generation. (2 marks)
- What does the 5th generation focus on? (2 marks)
- Differentiate between 1st and 2nd generation computers. (3 marks)

- What is a microprocessor, and which generation introduced it? (3 marks)

11. Long Answer Model Answers (5, 10 & 15 Marks)

Q1 (5 Marks): Explain any three generations of computers with their technology and examples.

Computer history is commonly divided into five generations, each defined by the main electronic technology used to build computers during that period. This answer explains three of these generations in detail.

The First Generation (1940-1956) used vacuum tubes as its core electronic component. Computers of this era, such as ENIAC, were extremely large, often filling an entire room, consumed huge amounts of electrical power, and generated significant heat, while also being highly unreliable due to the fragile nature of vacuum tubes.

The Third Generation (1964-1971) was defined by the Integrated Circuit (IC), a single chip containing many transistors combined together. This allowed computers, such as the IBM System/360, to become significantly smaller, faster, and more affordable than earlier generations, making them practical for use by ordinary businesses and universities for the first time.

The Fourth Generation (1971-present) introduced the microprocessor, an entire CPU manufactured onto a single chip. This breakthrough made personal computers, and eventually laptops and smartphones, possible — nearly every computer in everyday use today, including smartphones, belongs to this generation.

Together, these three generations illustrate a clear and consistent trend: with each advance in core technology, computers became progressively smaller, faster, and more accessible to ordinary people and organisations.

Q2 (10 Marks): Explain the generations of computers in detail, covering technology, size, speed, and example machines for each generation.

The history of electronic computers is commonly divided into five generations, each distinguished by the primary electronic technology used to build the computer's core processing circuitry, with a consistent trend of computers becoming smaller, faster, more reliable, and less expensive with each successive generation.

The First Generation (1940-1956) relied on vacuum tubes, fragile glass-enclosed switching devices similar in appearance to light bulbs. Computers of this era, such as ENIAC, which weighed about 30 tons and contained roughly 18,000 vacuum tubes, were room-sized, consumed enormous power, produced significant heat, and were highly unreliable, as tubes frequently burned out and needed constant replacement.

The Second Generation (1956-1963) replaced vacuum tubes with transistors — small, robust, and far more energy-efficient switching devices. This allowed computers such as the IBM 1401 to become considerably smaller, faster, and more dependable, similar to how bulky valve-based radios were replaced by small transistor radios during the same broad era.

The Third Generation (1964-1971) introduced the Integrated Circuit (IC), a single chip combining many transistors together. Computers such as the IBM System/360 became desk-sized, considerably faster, and more affordable, making them practical for widespread adoption by businesses, banks, and universities for the first time.

The Fourth Generation (1971-present) was defined by the microprocessor, which placed an entire CPU onto a single chip. This breakthrough made small, affordable Personal Computers possible, and this same underlying technology continued to shrink and improve over subsequent decades, eventually

enabling laptops and smartphones — nearly all computers in everyday use today, including the phone in a student's pocket, belong to this generation.

The Fifth Generation (present-future) is centred on Artificial Intelligence — computing systems capable of recognising speech, interpreting images, identifying complex patterns, and learning from data to improve their own performance. Voice assistants, face-unlock systems, and AI chatbots are everyday examples of this generation's capabilities. It is important to note that a device only qualifies as 5th generation if it is built around genuine AI capability — simply being newly manufactured does not qualify a device for this classification, and most current laptops remain 4th-generation, microprocessor-based machines.

- (A diagram showing all 5 generations with their shrinking relative sizes and years, as in Fig 3.1 of these notes, should be drawn here to earn full diagram marks.)

In conclusion, the five generations of computers together tell a remarkable story of continuous technological improvement, in which each new core technology — vacuum tubes, transistors, integrated circuits, microprocessors, and now artificial intelligence — enabled computers to become smaller, faster, cheaper, and more capable than the generation before it.

Q3 (15 Marks): Discuss, in detail, all five generations of computers, explaining the technology, key characteristics, advantages, limitations, and example machines of each, and explain the overall trend across all generations.

Computer history is conventionally divided into five distinct generations, each characterised by the dominant electronic technology used to construct computer systems during that era. Studying these generations in sequence reveals a remarkable and consistent pattern of technological progress that has shaped the modern digital world.

The First Generation (1940-1956) was built around vacuum tubes, fragile glass-enclosed electronic switching devices. Machines of this era, most famously ENIAC, completed in 1945, were extraordinarily large, often occupying entire rooms, weighing many tons, and containing thousands of individual vacuum tubes. These computers consumed massive amounts of electrical power, generated intense heat requiring dedicated cooling, and suffered from poor reliability, as vacuum tubes frequently failed and needed constant replacement by dedicated engineering teams. Programming was also extremely cumbersome, often requiring physical rewiring for each new task.

The Second Generation (1956-1963) marked a major improvement through the adoption of the transistor, a small, robust, and energy-efficient replacement for the vacuum tube. Computers such as the IBM 1401 became noticeably smaller, faster, more reliable, and less power-hungry, much like how bulky valve radios were being replaced by compact transistor radios during roughly the same period. This generation also saw computers become somewhat more commercially viable for large organisations to adopt.

The Third Generation (1964-1971) introduced the Integrated Circuit (IC), which allowed many individual transistors to be manufactured together onto a single small silicon chip. This further reduced the size and cost of computers while significantly improving speed and reliability. Machines such as the IBM System/360 became desk-sized and practical for widespread business, banking, and academic use, marking the point at which computing began moving beyond the exclusive domain of large government and research institutions.

The Fourth Generation (1971-present) was defined by arguably the most transformative single innovation in computing history: the microprocessor, which placed an entire Central Processing Unit onto one tiny chip. This breakthrough enabled the creation of small, genuinely affordable Personal Computers, and the same underlying technology continued to advance over subsequent decades, eventually giving rise to laptops and, ultimately, smartphones. The overwhelming majority of computing

devices in everyday use today — including laboratory desktop computers and the smartphone carried by nearly every student — belong to this generation.

- (A diagram showing the 5 generations with shrinking size and increasing speed, as in Fig 3.1 of these notes, should be included here for full diagram marks.)

The Fifth Generation (present-future) represents a shift in focus from simply improving raw hardware speed and size towards developing genuine Artificial Intelligence capability — systems that can recognise human speech, interpret images, identify complex patterns within large datasets, and progressively learn and improve from experience, rather than merely executing a completely fixed program. Everyday examples include voice assistants, face-unlock security systems, and AI-driven chatbots. It is important to clarify that classification as 'fifth generation' depends specifically on genuine AI capability, not merely on the age or novelty of a device — most current laptops and desktops remain fundamentally fourth-generation, microprocessor-based machines despite being recently manufactured.

Considered together, these five generations reveal an unmistakable and continuous overall trend: with every technological leap — from vacuum tubes, to transistors, to integrated circuits, to microprocessors, and now to artificial intelligence — computers have consistently become smaller in physical size, faster in processing speed, lower in manufacturing and purchase cost, more energy-efficient, and more broadly accessible to ordinary individuals and organisations. This sustained, multi-generational trend of miniaturisation and performance improvement is one of the most significant technological stories of the modern era, and it directly explains how computing evolved from a room-sized machine accessible only to governments and large corporations into the smartphone that virtually every student now carries in their pocket.

12. Exam Tips

- Memory trick: V-T-I-M-A — Vacuum tubes, Transistors, ICs, Microprocessors, Artificial Intelligence.
- Always give the correct year range along with the generation name — examiners specifically award marks for accurate dates.
- For 10 and 15 mark questions, always include the full comparison table AND the diagram (Fig 3.1) — both earn dedicated marks.
- Common Mistake: Do not confuse “newly purchased computer” with “5th generation computer” — the classification is based on the underlying AI technology, not the purchase date.

References

- **Refer to your university-prescribed/standard Computer Fundamentals textbook for this topic, as per syllabus official reading list.**
- **Class lecture discussion and board work for this lecture.**